

VLM-DexSafeDagger: VLM-Guided Intervention Policy for Safety-Critical Visuomotor Skill Distillation

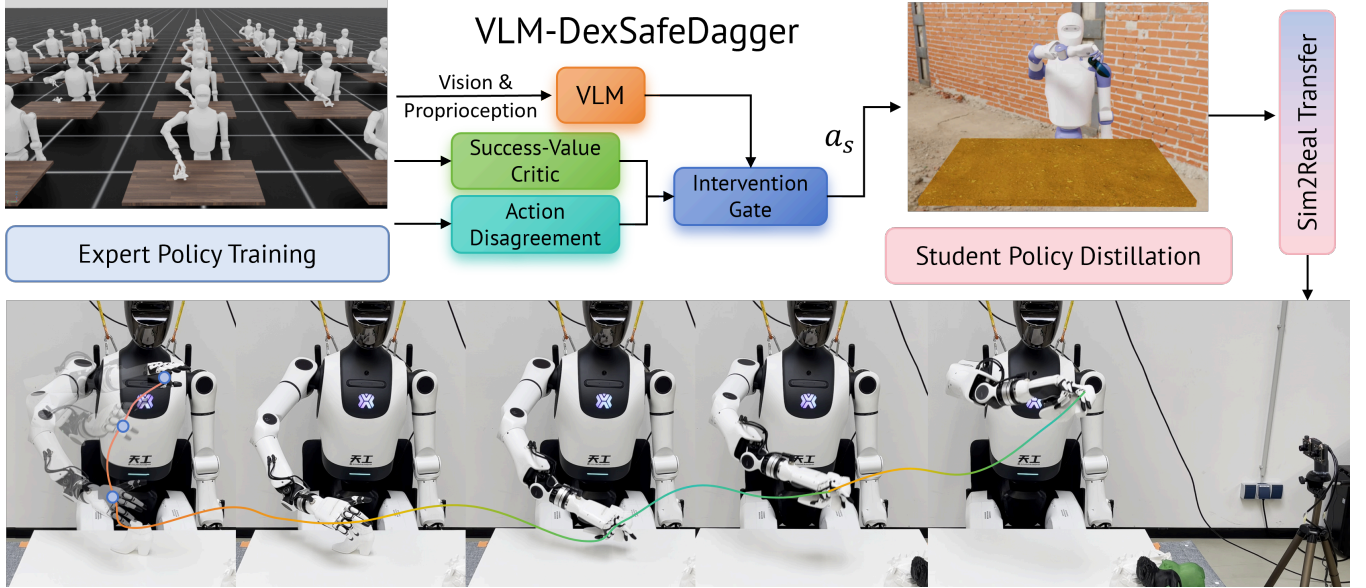


Fig. 1: **VLM-DexSafeDagger overview.** We propose a teacher-intervened online distillation pipeline that condenses teacher skills into a deployable RGB-based student policy. The unsafe exploration rate is substantially suppressed in the early training phase via a proposed VLM-guided intervention policy that combines teacher–student action disagreement with a Bellman success-value critic trained on safe-lift-success labels, while a vision-language model maintains an evolving, safety-aware understanding of the scene and periodically adjusts the intervention policy during online distillation.

Abstract—Learning dexterous manipulation is challenging due to high-dimensional action spaces and complex contact dynamics. In addition, safety-unaware exploration during learning can lead to hardware damage and unstable or even hazardous contact events, posing a significant barrier to real-world deployment. To address this challenge, we propose VLM-DexSafeDagger, a safety-aware online skill-distillation framework that transfers reinforcement learning (RL) teacher policies to a vision-based student policy. VLM-DexSafeDagger combines an ℓ_2 teacher–student disagreement gate with a learned Bellman success-value critic that estimates the prospective safe-lift-success value of the current student state–action pair. A vision-language model periodically adjusts the intervention policy parameterized by disagreement limit and acceptable predicted success rate. This design enables safer exploration during online interactive imitation learning using high-level visual context. Experiments on grasp-and-lift tasks show that VLM-DexSafeDagger significantly improves training safety, reducing unsafe episodes by up to 80% during the training period compared with vanilla DAgger-based distillation. Meanwhile, the distilled student policy achieves comparable lift success rates and demonstrates sufficient robustness for real-world robotic deployment.

I. INTRODUCTION

Dexterous robotic hands typically possess a large number of degrees of freedom (DoFs) and exhibit complex contact

dynamics, enabling them to perform fine-grained and diverse manipulation behaviors, such as multi-finger grasping [1], in-hand reorientation [2], [3], and precise control [4]. However, the high-dimensional action space and intricate interaction dynamics substantially increase the difficulty of control and policy learning. In recent years, reinforcement learning (RL)-based approaches [3], [5]–[7] have achieved remarkable progress in dexterous grasping and manipulation tasks. Leveraging large-scale parallel simulation and high-fidelity physics engines, RL enables the automatic learning of complex control policies in simulated environments without the need for manually designed controllers. These methods have demonstrated strong generalization capabilities across multi-object and multi-task scenarios.

Despite these advances, existing RL frameworks primarily optimize for task success rates and rarely incorporate explicit modeling or constraints related to safety-critical behaviors. During training, common failure modes include severe collisions between the hand and the environment, unintended object ejection due to excessive contact forces, and loss of stable hand posture [8], [9]. Such unsafe behaviors not only reduce training efficiency but also introduce potential risks during real-world deployment. Therefore, how to reduce

unsafe exploration while maintaining learning efficiency remains a critical challenge in dexterous manipulation.

To mitigate the performance gap between simulation and real-world deployment, skill distillation has become a widely adopted paradigm in recent years. The core idea is to first train a high-performing expert policy in simulation using privileged state information, and then distill its decision-making behavior into a student policy that relies solely on visual observations or partially observable inputs. This process enables modality transfer from privileged state information to perception-based inputs, facilitating efficient real-world deployment. Skill distillation reduces reliance on accurate state estimation and allows knowledge integration from multiple teacher policies, thereby substantially improving execution stability and generalization performance in real-world environments. Consequently, it has gained increasing attention in dexterous grasping and complex manipulation tasks. However, existing distillation methods [10], [11] typically rely on behavior cloning or the standard DAgger framework for data aggregation. Their primary objective is to mitigate performance degradation caused by distribution shift, rather than explicitly enforcing safety constraints during student policy learning. During distillation, the student policy may deviate from the teacher’s state-action distribution and enter regions that are either insufficiently covered by the teacher or inherently high-risk, leading to unsafe behaviors such as severe collisions or unstable hand configurations. Therefore, introducing explicit safety mechanisms into the distillation process—so that high-risk exploration can be avoided without sacrificing learning efficiency—remains an important and largely underexplored problem. Existing safety-aware distillation approaches, such as SafeDAgger [12], have primarily focused on autonomous driving scenarios and have not been systematically studied in the context of dexterous manipulation.

The contributions of this work are summarized as follows:

- We propose VLM-DexSafeDAgger, a VLM-guided, safety-sensitive framework for distilling RL-acquired teacher policies into a single RGB-based student policy. To the best of our knowledge, this is the first attempt in dexterous manipulation to incorporate a mixed intervention policy combining teacher-student action disagreement with a Bellman success-value critic, while the VLM reasons about scene-level safety from rollout statistics and representative visual observations to regulate the intervention gate.
- We propose a safety-aware RL reward design that explicitly accounts for risky events in arm-hand control scenarios, including harmful collisions, object escape, palm flipping, and the hand moving too far from the workspace, thereby encouraging safer exploration during teacher training.
- We empirically demonstrate that VLM-DexSafeDAgger, which combines disagreement, success-value prediction, and VLM-guided intervention policy adaptation, substantially reduces unsafe exploration compared with Vanilla DAgger, achieving an approximately **80%** rel-

ative reduction in the unsafe-episode rate during early training. It also reduces the average unsafe-episode rate by **7.5 percentage points** relative to disagreement-only SafeDAgger [12]. Ablation results further demonstrate the effectiveness of VLM-guided adaptation across different visual contexts and unsafe scenarios: VLM-DexSafeDAgger consistently achieves lower unsafe-episode rates than DexSafeDAgger, with an average reduction of **4 percentage points**.

II. RELATED WORK

Skill Distillation. Skill distillation transfers behavior from a high-capacity teacher to a compact student and has been widely adopted in both computer vision and robotics. In robotics, the teacher can leverage privileged state and simulator access during training, while the student is distilled to operate from on-board observations (e.g., RGB). DexTrAH-RGB [11] distills a privileged RL teacher into an RGB-based policy via DAgger-based imitation learning, while UniGraspTransformer [13] distills offline trajectories from object-specific teachers into a unified Transformer policy.

Interactive Imitation Learning. Many imitation learning methods operate offline, using a fixed dataset of expert demonstrations to match the expert action distribution [14], [15]. In sequential decision-making, however, distributional shift induces covariate shift and compounding errors. Interactive imitation learning [16] mitigates this issue by collecting supervision online along trajectories visited by the learner.

DAgger [10] addresses covariate shift by iteratively rolling out the learner and relabeling visited states with an expert, thereby aggregating data under the learner-induced state distribution. SafeDAgger [12] extends DAgger with a teacher-intervened rollout policy that takes over in high-risk states to reduce unsafe exploration. Recent variants [17], [18] extend interactive imitation learning to robotic manipulation, yet often rely on human supervision. Fully autonomous expert relabeling during online distillation remains rare in robotic manipulation, especially for dexterous, contact-rich tasks.

Vision-Language-Guided Robot Learning. Visual and semantic information is widely recognized as a strong cue for embodied agents to interpret task specifications and reason about unstructured environments [19], [20]. One line of work uses VLMs to generate demonstrations or supervisory signals for offline policy construction, such as Manipulate-Anything [21] and AHA [22]. Complementary approaches, including RACER [23] and FOREWARN [24], employ VLMs to guide risk prediction, provide recovery guidance for unsafe situations, or govern action selection during online policy learning or real-time execution.

III. PROBLEM STATEMENT AND METHODS

A. Problem Formulation

We consider online interactive imitation learning with teacher intervention in a partially observable Markov decision process (POMDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{O}, P, \Omega, r, \gamma)$, with privileged teacher state $s_t \in \mathcal{S}$, deployable student observation $o_t \in \mathcal{O}$ (RGB), action $a_t \in \mathcal{A}$, observation model $o_t \sim \Omega(\cdot \mid s_t)$,

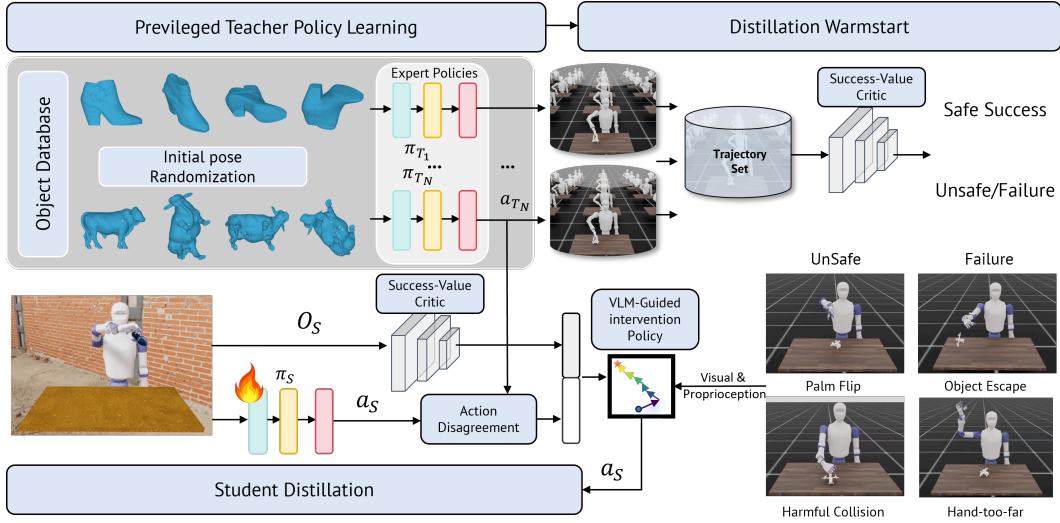


Fig. 2: Overview of the VLM-DexSafeDagger teacher-student learning pipeline. A privileged reinforcement learning teacher is trained in simulation and subsequently distilled into an RGB-based visuomotor policy for real-world deployment. A vision-language model periodically proposes a displacement of the intervention policy’s operating point; deterministic clipping and smoothing constrain the resulting policy update.

and dynamics $s_{t+1} \sim P(\cdot | s_t, a_t)$. During training, we assume access to a set of privileged teacher policies $\{\pi_{T_i}\}_{i=1}^K$, whereas at deployment the student policy π_S must act solely from o_t . Our goal is to distill the collective teacher behavior into an RGB-based student that achieves high task success while minimizing unsafe exploration during training.

The induced rollout distribution p_{π_G} is defined by the gated rollout policy π_G , which switches between the teacher and student policies according to the intervention gate. Its learning objective and intervention gate are

$$\begin{aligned} \min_{\pi_S \in \Pi_S} \max_{\mathcal{G} \in \mathcal{G}} & \left[\underbrace{-\mathbb{E}_{\tau \sim p_{\pi_G}} [\mathbb{I}_{\text{succ}}(\tau)]}_{\text{maximize success}} + \lambda \underbrace{\mathbb{E}_{\tau \sim p_{\pi_G}} \left[\sum_{t=0}^{T-1} \mathbb{I}_{\text{unsafe}}(s_t, a_t) \right]}_{\text{minimize unsafe}} \right] \\ \text{s.t. } & a_t = g_t a_t^T + (1 - g_t) a_t^S, \\ & g_t = \mathcal{G}(s_t, o_t; \tau^{(k)}) \in \{0, 1\}, \\ & s_{t+1} \sim P(\cdot | s_t, a_t), \quad o_t \sim \Omega(\cdot | s_t). \end{aligned} \quad (1)$$

where $g_t = 1$ indicates teacher takeover and $g_t = 0$ denotes student control. The intervention policy is parameterized by $\tau^{(k)} = (\tau_\delta^{(k)}, \tau_Q^{(k)})$, which governs teacher takeover decisions in $\mathcal{G}$. Periodically, a VLM proposes a displacement $\Delta\tau^{(k)}$, from which the policy is updated to adjust to distributional shift.

We formalize task success via a trajectory-level predicate $\mathbb{I}_{\text{succ}}(\tau) \in \{0, 1\}$, which is 1 if the object is grasped, lifted, and held above a height threshold for a consecutive duration. Safety is governed by early termination conditions: letting $\mathbb{I}_{\text{unsafe}}(s_t, a_t) \in \{0, 1\}$ indicate whether any unsafe event (e.g., harmful collision, object escape, palm flip, hand-too-far) occurs at time t , the trajectory is safe if $\prod_{t=0}^{T-1} (1 - \mathbb{I}_{\text{unsafe}}(s_t)) = 1$.

B. Overview of the Distillation Pipeline

Fig. 2 summarizes VLM-DexSafeDagger, our teacher-student distillation pipeline, which consists of five core components: (i) privileged teacher policy training in simulation (Sec. C), (ii) Bellman success-value critic learning to estimate safe lift success under student actions (Sec. D), (iii) deterministic teacher intervention (Sec. E), (iv) VLM-guided intervention policy adaptation (Sec. F), and (v) online student skill distillation (Sec. G).

The privileged teacher is first trained in simulation, and its rollouts provide an offline warm start for the Bellman critic. The critic estimates prospective lift success under the student action and continues to be updated from a ring replay buffer.

During online distillation, the teacher intervenes when either teacher-student action disagreement is high or the critic predicts low success. The student is then optimized online from teacher labels collected under the resulting rollout policy π_G . A VLM is periodically queried to propose updates to the intervention policy’s operating point. The resulting candidate operating point is clipped and smoothed before use. Pseudo-code is provided in Alg. 1.

C. Privileged Teacher Policy Training

We train each object-specific teacher π_T in simulation with PPO using an asymmetric actor-critic. The actor maps noisy, partial proprioceptive observations to robot actions, approximating the sensing available during rollout. The critic receives the corresponding clean, privileged robot-object state available only in simulation.

The reward shaping follows the below law

$$\begin{aligned} r_t &= \sum_{i \in \mathcal{R}} w_i R_{i,t} - \sum_{j \in \mathcal{P}} w_j P_{j,t}, \\ \mathcal{R} &= \{\text{dist, curl, align, contact, near, goal, lift}\}, \\ \mathcal{P} &= \{\text{action}\}. \end{aligned} \quad (2)$$

Algorithm 1 VLM-DexSafeDagger

```
1: Train the privileged teacher policy  $\pi_T$  via RL
2: Collect a warm-start dataset  $\mathcal{D}_0$  under teacher control
3: Pretrain the Bellman critic  $\hat{Q}_{\text{succ}}$  on  $\mathcal{D}_0$ 
4: Initialize the intervention policy parameters  $\tau^{(0)} = (\tau_\delta^{(0)}, \tau_Q^{(0)})$ 

5: Initialize the visual buffer  $\mathcal{B}_V$ 
6: for iteration  $k = 1 \dots N$  do
7:   Observe the teacher state  $s_t$  and student observation  $o_t$ 
8:   Compute the actions  $a_t^T \sim \pi_T(\cdot | s_t)$  and  $a_t^S \sim \pi_S(\cdot | o_t)$ 
9:   Compute the action disagreement  $\delta_t = \|a_t^S - a_t^T\|_2$ 
10:  Predict the safe-lift-success value  $\hat{Q}_{\text{succ}}(s_t, a_t^S)$ 
11:  Evaluate the intervention gate
12:    $g_t \leftarrow \mathbb{I}[\delta_t > \tau_\delta \vee \hat{Q}_{\text{succ}}(s_t, a_t^S) < \tau_Q]$ 
13:  Execute the gated action  $a_t = g_t a_t^T + (1 - g_t) a_t^S$ 
14:  Store the transition in the critic replay buffer  $\mathcal{D}$ 
15:  Update  $\hat{Q}_{\text{succ}}$ ; if  $g_t = 1$ , update  $\pi_S$ 
16:  Add representative visual states to  $\mathcal{B}_V$ 
17:  if an intervention policy update is due then
18:    Summarize rollout context  $\mathcal{C}_k$  and sample images  $\mathcal{V}_k$ 
19:    Query the VLM for the shift  $\Delta\tau^{(k)}$  given  $(\mathcal{C}_k, \mathcal{V}_k)$ 
20:    Form the candidate  $\tilde{\tau}^{(k)} = \tau^{(k)} + \Delta\tau^{(k)}$ 
21:    Clip and smooth  $\tilde{\tau}^{(k)}$  to obtain  $\tau^{(k+1)}$ 
22:  end if
23: end for
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The distance term guides the hand toward the object, while the curl and alignment terms encourage a grasp-ready, palm-down posture. The contact and stay-close terms promote stable thumb–finger contact and keep the hand near the object after reaching. The goal and lift terms then drive the grasped object toward the target and reward a stable lift. The action penalty discourages abrupt command changes.

Each episode lasts at most $T_{\text{max}} = 10$ s and terminates early upon a palm flip, harmful collision, object escape, or hand-too-far event (see Sec. A). The same event predicates are used to characterize teacher safety and to supervise subsequent intervention decisions.

D. Bellman Success-Value Critic

ℓ_2 action disagreement alone is insufficient as an intervention signal and may miss small action deviations that could cause unstable grasps, collisions, or object escape. We therefore learn a Bellman critic, trained with safe-lift-success labels, that estimates the probability of safe lift success for the current student state–action pair and provides a prospective signal complementary to action disagreement.

The Bellman critic evaluates a multimodal state–action pair (s_t, a_t^S) , where s_t combines robot proprioception and embeddings from a visual encoder. The encoder is trained end-to-end during skill distillation and regularized by the auxiliary object-position prediction loss in Sec. G. The twin critics Q_1 and Q_2 are ReLU MLPs with hidden dimensions [256, 128]. Each critic’s logit $z_i(s, a)$ is mapped to a success probability by $Q_i(s, a) = \sigma(z_i(s, a)/T)$, with calibration temperature $T = 4$. We use the conservative estimate $\hat{Q}_{\text{succ}}(s, a) = \min_{i \in \{1, 2\}} Q_i(s, a)$; lower values indicate higher prospective risk.

The critics are pretrained on teacher transitions and updated online from a ring replay buffer. The label $u_t \in \{0, 1\}$

indicates safe lift success: it is 1 only when the lift criterion is satisfied without an unsafe event. Training uses the SARSA-style target

$$y_t = \text{clip}\left(u_t + \gamma(1 - d_t^{\text{eff}}) \min_{i \in \{1, 2\}} Q_i^{\text{targ}}(s_{t+1}, a_{t+1}), 0, 1\right), \quad (3)$$

where d_t is the termination flag, h_t indicates whether a_{t+1} is valid, $d_t^{\text{eff}} = \max(d_t, 1 - h_t)$, and Q_i^{targ} are slowly updated target networks. Both critics minimize the mean-squared error (MSE) between their predictions and y_t ; the minimum across critics reduces overestimation.

E. Intervention Policy

The intervention policy uses a deterministic two-channel gate. The first channel measures teacher–student disagreement, $\delta_t := \|a_t^S - a_t^T\|_2$, where $a_t^T \sim \pi_T(\cdot | s_t)$ and $a_t^S \sim \pi_S(\cdot | o_t)$. The second channel evaluates prospective task success through $\hat{Q}_{\text{succ}}(s_t, a_t^S)$. The intervention policy is parameterized by $\tau^{(k)} = (\tau_\delta^{(k)}, \tau_Q^{(k)})$, where $\tau_\delta^{(k)}$ is the maximum admissible disagreement and $\tau_Q^{(k)}$ is the minimum acceptable predicted success.

These parameters define the deterministic intervention gate:

$$g_t = \mathbb{I}[\hat{Q}_{\text{succ}}(s_t, a_t^S) < \tau_Q^{(k)} \vee \delta_t > \tau_\delta^{(k)}]. \quad (4)$$

Thus, either large disagreement or low predicted success triggers teacher takeover. Decreasing τ_δ or increasing τ_Q makes the intervention policy more conservative; the opposite shift increases student autonomy.

We empirically initialize $\tau_\delta^{(0)}$ and $\tau_Q^{(0)}$ from the 90th percentile of warm-start disagreement and the 10th percentile of predicted success over successful teacher rollouts, respectively. These values define the initial operating point $\tau^{(0)}$.

F. VLM-Guided Intervention Policy Adaptation

During online distillation, the student policy, visited-state distribution, and Bellman critic’s calibration co-evolve, and the dominant failure modes may also change. A stationary intervention policy therefore cannot maintain a consistent safety–autonomy trade-off. We periodically query a VLM to guide the displacement of the intervention policy’s operating point. At iteration k , the VLM receives rollout statistics $\mathcal{C}_k$, including disagreement (δ_t), predicted success ($\hat{Q}_{\text{succ}}(s_t, a_t^S)$), intervention rate (β), and unsafe-episode rate (UnsafeRateAvg). It also receives a representative image batch $\mathcal{V}_k$ sampled from the visual buffer $\mathcal{B}_V$ to cover high-disagreement, low-success, and unsafe states. From this context, it proposes an operating-point displacement $\Delta\tau^{(k)} = (\Delta\tau_\delta^{(k)}, \Delta\tau_Q^{(k)})$, producing the candidate operating point $\tilde{\tau}^{(k)} = \tau^{(k)} + \Delta\tau^{(k)}$.

Each candidate policy parameter is clipped around its initial value and exponentially smoothed:

$$\tau_x^{(k+1)} = (1 - \alpha)\tau_x^{(k)} + \alpha \text{clip}\left(\tilde{\tau}_x^{(k)}, c_x^{\min}\tau_x^{(0)}, c_x^{\max}\tau_x^{(0)}\right), \quad x \in \{\delta, Q\}. \quad (5)$$

Here, α controls the adaptation rate, and $c_x^{\min}$ and $c_x^{\max}$ define the admissible calibration region around $\tau_x^{(0)}$.

G. Student Skill Distillation

Student learning is posed as supervised imitation from teacher-labeled transitions collected under the gated rollout policy π_G . The student observes RGB images (two stereo views) together with robot proprioception.

We optimize a combined objective $\mathcal{L} = \mathcal{L}_{\text{action}} + \mathcal{L}_{\text{aux}}$, where the auxiliary loss predicts object position to regularize the visual encoder, $\mathcal{L}_{\text{aux}} = \|\hat{\mathbf{x}}_{\text{obj}} - \mathbf{x}_{\text{obj}}\|_2$.

For action imitation, we use a weighted L_2 loss, $\mathcal{L}_{\text{action}} = \|\mathbf{a}_S - \mathbf{a}_T\|_W^2 = (\mathbf{a}_S - \mathbf{a}_T)^\top W (\mathbf{a}_S - \mathbf{a}_T)$, with $W = \text{diag}(w_1, \dots, w_d)$. We set $w_j = \sigma_{0,j}^{-2}$ using a reference scale $\sigma_{0,j}$ (e.g., the teacher action standard deviation on dimension j), yielding $\mathcal{L}_{\text{action}} = \sum_{j=1}^d (a_{S,j} - a_{T,j})^2 / \sigma_{0,j}^2$.

Training proceeds in two stages. (i) *Warm-start*: we collect an offline dataset (e.g., 100k transitions) under full teacher control ($g_t \equiv 1$) to pretrain the Bellman critic. (ii) *Online distillation*: we perform interactive rollouts under π_G and query teacher labels on intervened steps. The student is updated online using the current-step teacher label only; no replay buffer or offline dataset is maintained for student updating. The design reduces the effect of stale supervision and stabilizes online imitation updates.

IV. EXPERIMENT

A. Evaluation Metrics

For N evaluation episodes of 10 s, let τ index episodes and t timesteps.

a) *Lift success (ever-lifted with hold)*.

For object height h_t^τ , lift threshold H , and hold duration N_{hold} , define $\mathbb{I}_{\text{lift}}^\tau = \mathbb{I}[\exists t : \forall k \in \{0, \dots, N_{\text{hold}} - 1\}, h_{t+k}^\tau \geq H]$ and report $\text{LiftSucc} = \frac{1}{N} \sum_{\tau=1}^N \mathbb{I}_{\text{lift}}^\tau$.

b) *Unsafe episode rate*.

Let $\mathcal{M} = \{\text{coll}, \text{obj}, \text{flip}, \text{hf}\}$ denote harmful collision, object escape, palm flip, and hand-too-far, and let $\mathbb{I}_m^\tau$ indicate mode m in episode τ . Define $\mathbb{I}_{\text{unsafe}}^\tau = \mathbb{I}[\bigvee_{m \in \mathcal{M}} \mathbb{I}_m^\tau = 1]$ and report $\text{UnsafeRateAvg} = \frac{1}{N} \sum_{\tau=1}^N \mathbb{I}_{\text{unsafe}}^\tau$.

c) *Failure-mode decomposition*.

Each episode is assigned its first unsafe terminal mode $m^\tau \in \mathcal{M}$; a fixed priority resolves simultaneous triggers. We define $\mathbb{I}_m^\tau = \mathbb{I}[m^\tau = m]$ and report $\text{FailureModeRate}(m) = \frac{1}{N} \sum_{\tau=1}^N \mathbb{I}_m^\tau, m \in \mathcal{M}$.

B. Simulation Experiment Setup

Simulation experiments use single-arm control on a Tiangong 2 Pro humanoid equipped with InspireHand RH56DfX; the action space is 13-D (7D arm + 6D hand). The RGB student receives two 320×240 streams at 30 fps, encoded by ResNet-18 and fused into a 128-D actor feature. Training and distillation use Isaac Sim 5.0.0/Isaac Lab 2.2.1 on one RTX 5090 GPU.

We evaluate 40 objects from the Omniverse Asset Library and train one object-specific teacher for each object. Each teacher is trained for 12,000 episodes (3.5 h), totaling 480,000 teacher-training episodes and approximately 140 GPU-hours.

TABLE I: Teacher policy performance on four representative objects from the 40-object training set (200 evaluation episodes per object).

Metric (%)	WindCart	Boot	ToolBox	Cow
Lift Success	93.5	100.0	88.5	87.0
Unsafe Rate Avg	6.5	13.5	13.5	13.5
Harmful collision	0.0	0.0	6.5	0.0
Object escape	6.5	6.5	6.5	13.5
Palm flipped	0.0	6.5	0.0	0.0
Hand too far	0.0	0.0	0.0	0.0

Across the 40 teachers, lift-success rates range from 72–100%, and unsafe rates range from 5–30%; Table I reports a representative four-object subset. During distillation, we randomize lighting intensity, direction, background textures, object material, and table appearance following DexTrAH-RGB-style domain randomization. The joint student is distilled across all 40 objects for 100,000 episodes in 2.5 h.

We use GPT-5.5 with high reasoning effort as the VLM. We issue the first VLM query after 2,000 episodes and subsequently query it every 10,000 distillation episodes, resulting in 10 VLM calls over the full 100,000-episode training run and approximately 70,000 tokens of total usage per run.

C. Ablation Studies

The following ablation studies validate the design choices in VLM-DexSafeDagger. We distill the teacher policies trained during the RL phase into a compact, generalizable student policy. For RL training, we compare the standard DexTrAH-RGB pipeline with our customized variant. For skill distillation, we evaluate four methods—Vanilla DAgger, Vanilla SafeDagger, DexSafeDagger, and VLM-DexSafeDagger—in terms of the evolution of lift-success, unsafe-episode, and intervention rate throughout training. The intervention rate is defined as the fraction of time steps at which the teacher takes over: $\beta = \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{I}[g_t = 1]$. We average this metric across evaluation episodes. As our primary safety metric, we report the unsafe-episode rate averaged across the four ablation objects, together with breakdowns by object and failure mode. The ablation settings are defined below.

(a) **DexTrAH-RGB RL + Vanilla DAgger**. The teacher is trained using the DexTrAH-RGB RL recipe, and the student policy is trained using Vanilla DAgger. The student controls the rollout ($\beta = 0$), while the parallel teacher policy labels each executed student action a_t^S with the corresponding teacher action a_t^T .

(b) **Our RL + Vanilla DAgger**. This setting matches (a) except that the teacher is trained with our customized RL recipe instead of the DexTrAH-RGB pipeline.

(c) **Our RL + Vanilla SafeDagger**. Using the same teacher as in (b), we replace Vanilla DAgger with a disagreement-gated SafeDagger baseline. The teacher intervenes only when $\|a_t^T - a_t^S\|_2$ exceeds a fixed threshold, and we train the student on teacher-labeled pairs (o_t^S, a_t^T) .

(d) **DexTrAH-RGB RL + DexSafeDagger**. DexSafeDagger

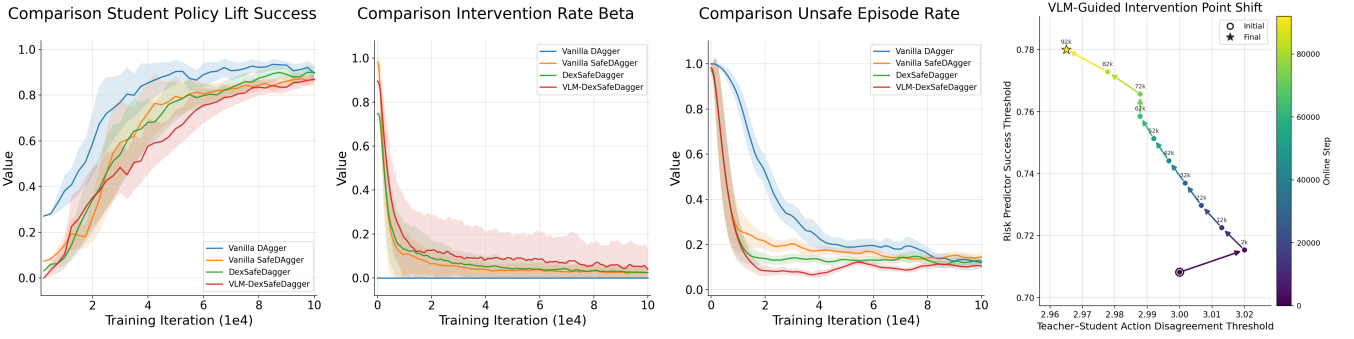


Fig. 3: **Ablation studies.** The first three panels compare four distillation variants in terms of student lift success, teacher intervention rate β , and unsafe episode rate, respectively, all plotted over training iterations. The rightmost panel shows the VLM-guided trajectory of the intervention operating point over online training. Shaded regions indicate variability across runs (mean $\pm$ std). Curves correspond to vanilla DAgger (blue), vanilla SafeDagger (orange), DexSafeDagger (green), and VLM-DexSafeDagger (red).

TABLE II: Performance of ablations (a)–(f) on four representative objects from the 40-object training set (200 evaluation episodes per object). Each ablation reports lift success and average unsafe rate.

Metric (%)	WindCart		Boot		ToolBox		Cow	
	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe
Ablation (a)	75.0	24.0	76.5	28.0	72.0	26.0	70.0	27.5
Ablation (b)	88.5	13.5	89.0	20.5	82.5	19.0	88.0	17.5
Ablation (c)	88.0	12.0	89.5	19.5	81.5	18.5	87.0	16.0
Ablation (d)	74.5	23.0	75.0	27.5	70.5	26.5	69.5	27.0
Ablation (e)	88.0	7.5	90.0	15.0	82.0	15.5	88.0	12.0
Ablation (f)	88.5	6.0	90.0	10.5	82.5	11.0	88.5	8.5

denotes SafeDagger with disagreement and a Bellman success-value critic under a stationary intervention policy. This setting uses the DexTrAH-RGB-trained teacher (as in (a)). Intervention is triggered by either large teacher–student disagreement or low predicted success, and the student is trained from teacher-labeled pairs (o_t^S, a_t^T) collected under this gate.

(e) Our RL + DexSafeDagger. This setting uses the DexSafeDagger setup defined in (d), but replaces the DexTrAH-RGB-trained teacher with one trained using our customized RL recipe (as in (b)).

(f) Our RL + VLM-DexSafeDagger. This setting extends (e) with the proposed VLM tuner, which periodically calibrates the disagreement-risk intervention policy using rollout statistics and representative visual states.

Table II shows that ablations (a) and (d), which use the DexTrAH-RGB RL recipe, achieve lower lift success and higher unsafe rates. We attribute this gap to the recipe’s reliance on a FABRICS controller tuned for the KUKA–Allegro platform, which transfers poorly to our hardware and dynamics. The resulting weaker teachers constrain downstream distillation performance.

As shown in Fig. 3, ablations (b), (c), (e), and (f) achieve comparable final lift success (left), consistent with the intervention policy’s focus on safety rather than task success. Unlike Vanilla DAgger ($\beta = 0$), the three intervention variants

reduce teacher intervention as the student improves (middle). They reduce early unsafe episodes by approximately 80% relative to Vanilla DAgger and remain below 20% throughout distillation, with VLM-DexSafeDagger achieving the lowest rate overall (right).

Comparing ablations (e) and (f) isolates the contribution of VLM-guided adaptation in VLM-DexSafeDagger. Both use the same RL teacher and two-channel intervention policy, but (e) uses fixed thresholds, whereas (f) adapts them using a VLM conditioned on rollout statistics and visual context. Across all evaluated objects, ablation (f) achieves an unsafe-episode rate that is lower than that of (e) by an average of 4 percentage points, while maintaining comparable lift success. This result shows that VLM-guided intervention policy adaptation improves safety without sacrificing task performance.

Compared with ablation (c), which uses disagreement alone, ablation (e) adds success-value prediction while retaining the same RL teacher and stationary intervention policy. The combined gate intervenes in low-predicted-success states missed by disagreement alone, reducing the average unsafe-episode rate by about 7.5 percentage points and limiting collision and object-escape failures without sacrificing task performance. Its unsafe rate approaches the teacher’s average ($\approx 12\%$) by the end of training and reaches a minimum of approximately 8%.

Fig. 4 shows that harmful collisions and object escape dominate unsafe episodes. Compared with Vanilla DAgger, all three intervention variants suppress both failure modes, particularly their early spikes, demonstrating that teacher intervention limits unsafe exploration during the initial distillation phase.

D. Real-World Experimental Results

We evaluate the deployed student policy on 22 objects, comprising 16 seen objects and 6 unseen objects. Table III reports a representative six-object subset. The experiments exhibit a lower lift success rate and a higher unsafe episode rate than the teacher and student evaluation results in

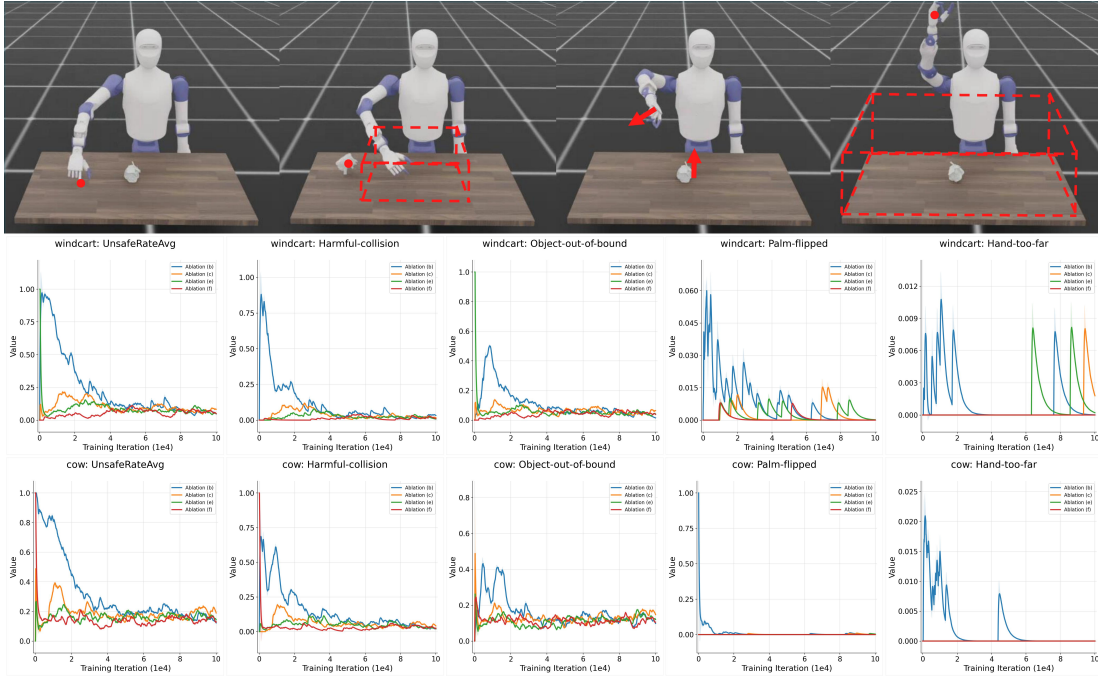


Fig. 4: **Per-object failure-mode decomposition during online distillation ablations.** We illustrate four task-specific failure modes in simulation (harmful collision, object escape, palm flipped, and hand too far) and report the aggregate unsafe rate together with its per-mode breakdown over training iterations. Results are shown for two objects (windcart and cow); lower is better in all plots. Curves correspond to four distillation variants: blue, Vanilla DAgger [10]; orange, Vanilla SafeDagger [12]; green, DexSafeDagger; and red, VLM-DexSafeDagger.

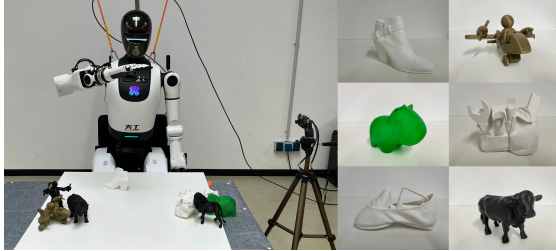


Fig. 5: **Real-world experimental setup and test objects.** Left: the single-arm manipulation platform used for deployment and evaluation. Right: examples of the physical objects used in the real-world lift experiments.

TABLE III: Representative six-object subset of the 22-object real-world evaluation (15 episodes per object). **Unseen objects** are shaded.

Metric (%)	WindCart	Boot	ToolBox	Cow	Plane	Shoe
Lift Success	86.7	86.7	80.0	86.7	66.7	60.0
Unsafe Rate Avg	13.3	20.0	20.0	26.7	40.0	40.0
Harmful collision	0.0	6.7	13.3	13.3	20.0	26.7
Object escape	13.3	13.3	6.7	13.3	20.0	13.3
Palm flipped	0.0	0.0	0.0	0.0	0.0	0.0
Hand too far	0.0	0.0	0.0	0.0	0.0	0.0

simulation, indicating that VLM-DexSafeDagger does not fully eliminate the sim-to-real gap. Nevertheless, the resulting policy is sufficiently robust for real-world deployment.

The unseen-object evaluation reveals a clear generalization gap: compared to seen objects (WindCart/Boot/ToolBox/Cow), the policy achieves lower lift success and higher unsafe rates on unseen objects (Plane/Shoe). This degradation is primarily driven by increased harmful collisions (20.0% and 26.7% on Plane and Shoe), while object escape remains comparable across objects (≈ 6.7 –20.0%). No palm-flip or hand-too-far events are observed, suggesting that real-world failures are dominated by contact-rich interactions rather than gross pose instabilities. Overall, the results indicate that the remaining sim-to-real gap on unseen geometries is largely attributable

to a contact-strategy mismatch, i.e., the policy fails to adapt its contact interactions to novel object shapes.

V. CONCLUSION AND FUTURE WORK

In this work, we propose VLM-DexSafeDagger, which integrates teacher–student disagreement with a Bellman success-value critic trained on safe-lift-success labels and uses a VLM to adapt the intervention policy. VLM-DexSafeDagger improves exploration safety during multi-teacher distillation, reducing the unsafe-episode rate by **80%** relative to the vanilla DAgger baseline during early training. The introduction of the Bellman critic reduces the average unsafe-episode rate by **7.5 percentage points** relative to the disagreement-only SafeDagger baseline. In addition, VLM-guided intervention policy adaptation yields consistent improvements across different visual contexts and unsafe scenarios, reducing the average unsafe-episode rate by a further **4 percentage points**

relative to DexSafeDagger. The proposed method approaches and, in some cases, surpasses the teacher policy’s safety performance. Successful real-world deployment shows that the resulting student policy partially bridges the sim-to-real gap.

A clear limitation is that performance during distillation, as well as the final student policy, is largely constrained by teacher capability. Developing a safety-aware teacher training pipeline is a promising direction for future work to generate stronger teacher policies.

APPENDIX

TRAINING AND INFERENCE DETAILS

A. RL Training Details

[Placeholder: Report the complete actor and critic observation vectors, including their dimensions, normalization, observation noise, and privileged inputs. Specify the actor-critic architecture and all PPO hyperparameters, including the optimizer, learning rate, discount factor, GAE parameter, clip range, entropy and value-loss coefficients, batch size, rollout length, number of environments, update schedule, and training duration. Provide the numerical weights and definitions of every reward and penalty term in Eq. (2), together with the initialization and domain randomization settings used for teacher training.]

B. VLM Inference Details

[Placeholder: Provide the complete system and user prompts, exact model and version identifier, access date, decoding and reasoning settings, and API configuration. Define every rollout statistic supplied to the VLM and specify the number of images, camera views, resolution, preprocessing, and image-selection procedure used for each query. Include the machine-readable output schema, parsing and validation logic, threshold ranges, clipping and smoothing rules, query schedule, retry and timeout behavior, malformed-output handling, and deterministic fallback. Report the measured latency, token usage, and monetary cost per call and per training run.]

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